

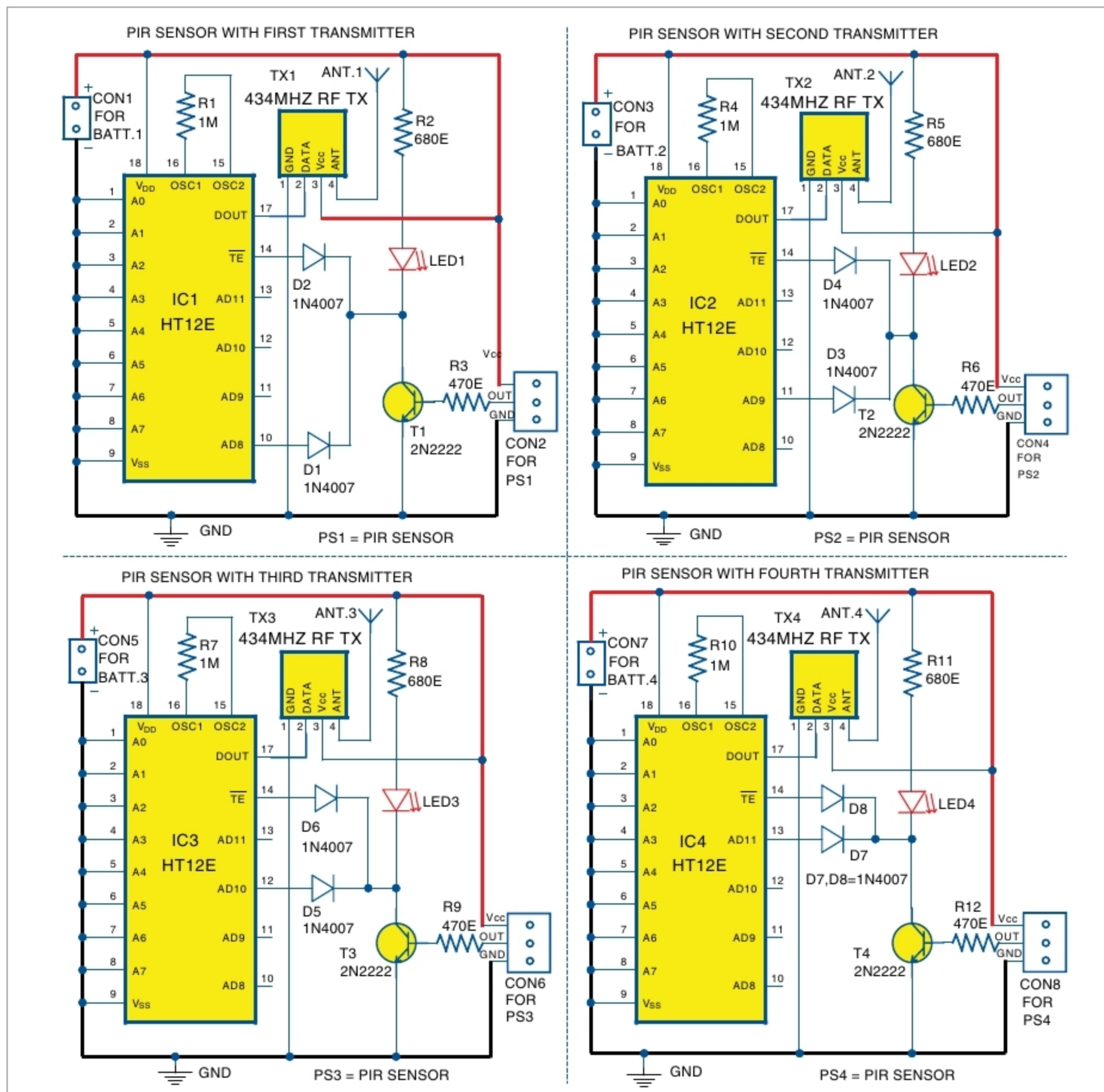


Fig. 3: The four transmitter circuits

LED at low frequency (1Hz - 2Hz), and the other for generating the audio frequency signal (1kHz) through the speaker for siren.

Circuit and working

There are five circuits. Of these, four transmitter circuits are similar to each other, with minor changes, as shown in Fig. 3. Fifth circuit, as shown in Fig. 4, is the receiver circuit.

Transmitter circuit. The first transmitter circuit is built around IC1.

Output of PIR sensor PS1 is given to data input AD8 (pin 10) of HT12E (IC1) after inverting it through transistor T1. Diodes D1 and D2 connect transmission enable ($\overline{TE}$) pin with pin AD8, so that both pins get input from the sensor at the same time.

LED1 is connected to collector output of T1, so that it blinks when PS1 sensor output goes high. Address pins A0 through A7 of IC1 are connected to ground to set address 00 (0000 0000b). Serial data output D_{OUT} (pin

17) is given as data input to 434MHz RF Tx module (TX1). The whole circuit is given power through 6V battery (BATT.1) connected across CON1.

The other three circuits shown in Fig. 3 are similar to the one described above. The only difference is that the diode is connected to different data pins.

When PIR sensor PS1 detects motion, its output goes high. T1 conducts and LED1 blinks. Pins AD8 and $\overline{TE}$ of IC1 are pulled low together